

Mathematical Cheatsheet: SBLToolkit.jl, C-Z0HR, and GSPT

1. Upstream C-Z0HR Regularization Pipeline & Soft-Plus Closure

- **Discretized Spatial Curvature Scale (C):**

$$C(z, \Delta z) = \frac{1}{2} \left| \frac{\partial^2 Ri_g}{\partial z^2} \right| (\Delta z)^2 + \epsilon_c Ri_c \quad (C > 0) \quad (1)$$

- C^∞ **Hyperbolic Distance Metric (Φ_ϵ):**

$$\Phi_\epsilon(x) = \sqrt{x^2 + \epsilon^2}, \quad x \equiv Ri_g - Ri_c \quad (2)$$

- **Upstream Regularized Richardson Number Mapping (Ri_g^{reg}):**

$$Ri_g^{\text{reg}} = Ri_c + (Ri_g - Ri_c) \left[\frac{\Phi_\epsilon(Ri_g - Ri_c) + C}{\Phi_\epsilon(Ri_g - Ri_c) + (1 + \alpha)C} \right] \quad (3)$$

- C^∞ **Hyperbolic Soft-Plus Evaluation:**

$$\text{softplus}_\epsilon(g) \equiv \frac{1}{2} \left(g + \sqrt{g^2 + \epsilon^2} \right), \quad g \equiv 1 - \frac{Ri_g^{\text{reg}}}{Ri_c} \quad (4)$$

- **Regularized Stability & Exchange Closures ($S_m^{\text{reg}}, K_{m,h}$):**

$$S_m^{\text{reg}}(Ri_g^{\text{reg}}) = \left[\text{softplus}_\epsilon \left(1 - \frac{Ri_g^{\text{reg}}}{Ri_c} \right) \right]^2 \quad (5)$$

$$K_m = l_m^2 S \cdot S_m^{\text{reg}}(Ri_g^{\text{reg}}), \quad K_h = l_h^2 S \cdot S_h^{\text{reg}}(Ri_g^{\text{reg}}) \quad (6)$$

2. Sensitivity Attenuation & Analytical Jacobians

- **Partial Mapping Slope ($\mathcal{D}_\alpha^{(R)}$):**

$$\mathcal{D}_\alpha^{(R)}(x) \equiv \left. \frac{\partial Ri_g^{\text{reg}}}{\partial Ri_g} \right|_C = \frac{\Phi_\epsilon(x) + C}{\Phi_\epsilon(x) + (1 + \alpha)C} + \frac{\alpha C x^2}{\Phi_\epsilon(x) [\Phi_\epsilon(x) + (1 + \alpha)C]^2} \quad (7)$$

- **Critical Threshold Attenuation ($x = 0$):**

$$\mathcal{D}_\alpha^{(R)}(0) = \frac{\epsilon + C}{\epsilon + (1 + \alpha)C} \xrightarrow{\epsilon \ll C} \frac{1}{1 + \alpha} \quad \left(\text{for } \alpha = 2.0, \mathcal{D}_2^{(R)}(0) = \frac{1}{3} \right) \quad (8)$$

- **Continuous Analytical Exchange Jacobian ($\frac{\partial S_m^{\text{reg}}}{\partial Ri_g}$):**

$$\frac{\partial S_m^{\text{reg}}}{\partial Ri_g} = -\frac{\text{softplus}_\epsilon(g)}{Ri_c} \left(1 + \frac{g}{\sqrt{g^2 + \epsilon^2}} \right) \mathcal{D}_\alpha^{(R)}(Ri_g) \quad (9)$$

3. Surface Energy Budget (SEB) & Discrete Gain Chain

- **Surface Energy Balance (SEB):**

$$C_s \frac{\partial T_s}{\partial t} = R_n(t) - H_0(t) - G_s(t) - LE_0(t) \quad (10)$$

$$H_0 = \rho c_p K_h(z_1) \left(\frac{\theta_1 - T_s}{z_1} \right), \quad G_s = K_s(T_s - T_{\text{deep}}) \quad (11)$$

- **Backward-Euler Implicit T_s^{n+1} Update:**

$$T_s^{n+1} = \frac{1}{\beta} \left[\frac{C_s}{\Delta t} T_s^n + R_n^{n+1} + \gamma \theta_1^{n+1} + K_s T_{\text{deep}} \right] \quad \text{where } \gamma \equiv \frac{\rho c_p K_h^{n+1}}{z_1}, \beta \equiv \frac{C_s}{\Delta t} + \gamma \quad (12)$$

- **Discrete Loop Gain Chain ($\mathcal{G}$):**

$$\mathcal{G} = \underbrace{\left(\frac{\partial Ri_g}{\partial \theta_z} \right)}_{\frac{q}{\theta_0 S^2}} \left(\frac{\partial \theta_z}{\partial H} \right) \left(\frac{\partial H}{\partial K_h} \right) \mathcal{K}'_h(Ri_g^{\text{reg}}) \cdot \mathcal{D}_\alpha^{(R)}(Ri_g) \quad (13)$$

- **Conditional Stability Bound:**

$$|\mathcal{G}_{\text{physical}}(Ri_c)| < 1 + \alpha \implies |\mathcal{G}_{\text{reg}}(Ri_c)| < 1 \quad (14)$$

4. GSPT Height-Resolved Curvature Decomposition

- **Richardson Profile Relation:**

$$Ri_g(z) = R(\zeta(z)), \quad \zeta(z) = \frac{z}{L(z)} \quad (15)$$

- **Exact Spatial Curvature Decomposition:**

$$\frac{d^2 Ri_g}{dz^2} = \underbrace{R''(\zeta) \zeta_z^2}_{C_{\text{const}}(z)} + \underbrace{R'(\zeta) \zeta_{zz}}_{C_{\text{coord}}(z)} + \mathcal{E}_{\Delta z} \quad (16)$$

- **Coordinate Jacobian (ζ_z) & Acceleration (ζ_{zz}):**

$$\zeta_z = \frac{L(z) - zL'(z)}{L(z)^2} = \frac{1 - \zeta L'(z)}{L(z)} \quad (17)$$

$$\zeta_{zz} = -\frac{zL''(z)}{L(z)^2} - \frac{2L'(z)}{L(z)} \zeta_z \quad (18)$$

- **Linear Obukhov No-Fold Test ($L(z) = L_0 + az$):**

$$L(z) = L_0 + az \quad (L_0 > 0, a \geq 0) \implies \zeta_z = \frac{L_0}{(L_0 + az)^2} > 0 \quad \forall z \geq 0 \quad (19)$$

- **Nondegenerate Coordinate Fold Condition ($\zeta_z = 0$):**

$$\zeta_z(z^*) = 0 \iff L(z^*) = z^* L'(z^*), \quad \zeta_{zz}(z^*) = -\frac{z^* L''(z^*)}{L(z^*)^2} \neq 0 \quad (20)$$

$$\left. \frac{d^2 Ri_g}{dz^2} \right|_{z^*} = R'(\zeta(z^*)) \zeta_{zz}(z^*) + \mathcal{E}_{\Delta z} \quad (\text{since } C_{\text{const}}(z^*) \equiv 0) \quad (21)$$

- **Inflection Masking Condition ($Ri_{zz} = 0$):**

$$(1 + \beta \zeta) \zeta_{zz} - 2\beta \zeta_z^2 = 0 \implies \frac{\zeta_{zz}}{\zeta_z^2} = \frac{2\beta}{1 + \beta \zeta} = \frac{2\beta L}{L + \beta z} \quad (22)$$

5. Fast–Slow Dynamical Systems & Bifurcated TKE Mechanics

- **Fast TKE Equation ($\dot{\epsilon}$):**

$$\epsilon \frac{de}{dt} = \mathcal{P} - \mathcal{B}(e) - \varepsilon_{\text{diss}} = l_0 \sqrt{e + \delta} S^2 - B_{0,\text{max}} \left(\frac{e^2}{e^2 + \delta_{\text{reg}}^2} \right) - \frac{e^{3/2}}{l_0 \left(1 + \frac{\beta N^2}{S_{\text{eff}}^2} \right)} \quad (23)$$

- **Slow Wind Shear Equation ($\dot{S}$):**

$$\frac{dS}{dt} = G_0 - \gamma_s e S - r_s S \quad (24)$$

- **Fast Stability Eigenvalue (λ_f):**

$$\lambda_f(e, S) = \frac{1}{\epsilon} \frac{\partial F}{\partial e} \quad (\text{Attracting sheet where } \lambda_f < 0) \quad (25)$$

- **Unregularized Saddle-Node Fold Locus:**

$$e_{\text{fold}} = \frac{l_0}{\sqrt{3}} \sqrt{S_{\text{fold}}^2 + \beta N^2}, \quad S^4(S^2 + \beta N^2) = \frac{27 B_{0,\text{max}}^2}{4 l_0^4} \quad (26)$$

- **Geometric–Dynamical Decoupling Principle:**

$$\underbrace{\zeta_z(z^*, t) = 0}_{\text{Kinematic Fold}} \not\iff \underbrace{\lambda_f(e, S) = 0}_{\text{Bifurcation}} \quad (27)$$

6. Track A Primitive Regularization & Non-Commutation Operators

- **Non-Commutation Operator Hazard:**

$$\mathcal{M}_\delta[A/B] \neq \mathcal{M}_\delta A / \mathcal{M}_\delta B \quad (28)$$

- **Tikhonov–Morozov Primitive Spline Functional ($\xi \equiv \ln(z/z_0)$):**

$$\min_{f_s} \sum_{i=1}^{N_{\text{obs}}} \left(\frac{f_s(\xi_i) - q_i}{\sigma_i} \right)^2 + \alpha \int_0^{\xi_{\text{max}}} [f_s''(\xi)]^2 d\xi \quad \text{s.t. } \|f_s(\xi_i) - q_i\|_2 = \delta \quad (29)$$

- **Tangential Cone Condition (Non-linear Convergence):**

$$\|F(x_2) - F(x_1) - F'(x_1)(x_2 - x_1)\|_Y \leq \gamma \|F(x_2) - F(x_1)\|_Y \quad (30)$$

- **Analytical Log-Coordinate Derivatives:**

$$\frac{df}{dz} = \frac{1}{z} \frac{df}{d\xi}, \quad \frac{d^2 f}{dz^2} = \frac{1}{z^2} \frac{d^2 f}{d\xi^2} - \frac{1}{z^2} \frac{df}{d\xi} \quad (31)$$

7. SCM Spatial Curvature Error Partitioning

- **Closed Spatial Curvature Residual Partition:**

$$\Delta_{\text{SCM}}(z) \equiv Ri_{zz}^{\text{SCM}}(z) - Ri_{zz}^{\text{obs}}(z) = \Delta_{\text{closure}}(z) + \Delta_{\text{geom}}(z) + \tau_{\Delta z}(z) \quad (32)$$

- **Discretization Truncation Residual ($\tau_{\Delta z}$):**

$$\tau_{\Delta z}(z) = \frac{(\Delta z)^2}{12} \frac{\partial^4 Ri_g}{\partial z^4} + \mathcal{O}((\Delta z)^4) \quad (33)$$